

Concept 2 | Polynomial Division

English Student Edition • Focused formulas and guided practice

Formula opener

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder

$$R = P(a) \quad \text{for divisor } x - a$$

Degree check

$$\deg R < \deg B$$

Classified Practice

Polynomial Division

Q001 Challenge

Find the quotient and the remainder when $2x^3 + x^2 - 7x + 3$ is divided by $x^2 + x - 2$. Then verify the result using $A = B \times Q + R$:

Choose the correct option.

A $Q = 2x - 2, \quad R = -2x + 2$

B $Q = 2x - 1, \quad R = -2x + 1$

C $Q = 2x - 1, \quad R = 2x - 1$

D $Q = 2x, \quad R = -2x + 1$

Classified Practice

Polynomial Division

Q002 Challenge

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

Choose the correct option.

A $3x^3 - 7x^2 + 12x - 1$

B $(3x - 1) \times (x^2 - 2x + 3)$

C $3x^3 - 7x^2 + 10x - 5$

D $x^2 + 2x + 4$

Classified Practice

Polynomial Division

Q003 Challenge

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

Choose the correct option.

A $\frac{x^3+5x^2+9x-11}{x+3}$

B $\frac{-x^2-2x+1}{x^3+5x^2+5x-3}$

C $\frac{x^3+5x^2+5x-3}{x+4}$

D $x^2 + 2x - 1$

Classified Practice

Polynomial Division

Q004 Challenge

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x . State the mathematical check Which of the following is correct?:

Choose the correct option.

A $2x^3 + 10x^2 + 6x - 18$

B $2x + 6$

C $2x + 5$

D $\frac{1}{2x+5}$

Classified Practice

Polynomial Division

Q005 Challenge

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$. Decide whether the result is exact and justify from R:

Choose the correct option.

A $Q = 3x^2 + 5x + 5, \quad R = -2x - 10$

B $Q = 3x^2 + 5x + 6, \quad R = -2x - 11$

C $Q = 3x^2 + 5x + 6, \quad R = 2x + 11$

D $Q = 3x^2 + 5x + 7, \quad R = -2x - 11$

Classified Practice

Polynomial Division

Q006 Challenge

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

Choose the correct option.

A $Q = x \times (x^2 - 2x + 1), \quad R = 6$

B $Q = x \times (x^2 - 2x + 1), \quad R = -6$

C $Q = x^3 - 2x^2 + x + 1, \quad R = 6$

D $Q = x^3 - 2x^2 + x - 1, \quad R = 7$

Classified Practice

Polynomial Division

Q007 Written

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

Q008 Written**Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.**

Classified Practice

Polynomial Division

Q009 MCQ

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:

Choose the correct option.

A $Q = 3x^2 + 8x + 9, \quad R = -23$

B $Q = 3x^2 + 8x + 10, \quad R = 23$

C $Q = 3x^2 + 8x + 8, \quad R = 24$

D $Q = 3x^2 + 8x + 9, \quad R = 23$

Classified Practice

Polynomial Division

Q010 MCQ

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:

Choose the correct option.

A $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{37}{8}, R = \frac{111}{8}$

B $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, R = \frac{103}{8}$

C $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, R = -\frac{103}{8}$

D $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{53}{8}, R = \frac{103}{8}$

Classified Practice

Polynomial Division

Q011 MCQ

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:

Choose the correct option.

A $Q = x \times (3x - 1), \quad R = 9$

B $Q = 3x^2 - x + 1, \quad R = 8$

C $Q = 3x^2 - x + 1, \quad R = -8$

D $Q = 3x^2 - x + 2, \quad R = 8$

Q012 MCQ

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:

Choose the correct option.

A $Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$

B $Q = 2x^3 - 3x^2 + 6x - 17, \quad R = -38$

C $Q = 2x^3 - 3x^2 + 6x - 16, \quad R = 38$

D $Q = 2x^3 - 3x^2 + 6x - 18, \quad R = 39$

Classified Practice

Polynomial Division

Q013 MCQ

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:

Choose the correct option.

A $Q = 4x^3 + 5x^2 + 10x + 21, \quad R = -36$

B $Q = 4x^3 + 5x^2 + 10x + 22, \quad R = 36$

C $Q = 4x^3 + 5x^2 + 10x + 20, \quad R = 37$

D $Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$

Classified Practice

Polynomial Division

Q014 MCQ

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:

Choose the correct option.

A $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{100}{27}, \quad R = -\frac{92}{27}$

B $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$

C $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = \frac{119}{27}$

D $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{46}{27}, \quad R = -\frac{119}{27}$

Classified Practice

Polynomial Division

Q015 Written

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

Q016 Written**Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.**

Classified Practice

Polynomial Division

Q017 Written

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

Q018 Written**Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.**

Classified Practice

Polynomial Division

Q019 MCQ

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:

Choose the correct option.

A $Q = x + 1, \quad R = 4x - 6$

B $Q = x + 2, \quad R = -4x + 6$

C $Q = x, \quad R = -4x + 7$

D $Q = x + 1, \quad R = -4x + 6$

Q020 MCQ

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:

Choose the correct option.

A $Q = x^2 - 5x + 10, \quad R = -25x + 7$

B $Q = x^2 - 5x + 11, \quad R = -25x + 6$

C $Q = x^2 - 5x + 11, \quad R = 25x - 6$

D $Q = x^2 - 5x + 12, \quad R = -25x + 6$

Classified Practice

Polynomial Division

Q021 MCQ

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:

Choose the correct option.

A $Q = 2x + 5, \quad R = -2x + 5$

B $Q = 2x + 6, \quad R = 2x - 5$

C $Q = 2x + 4, \quad R = 2x - 4$

D $Q = 2x + 5, \quad R = 2x - 5$

Q022 MCQ

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:

Choose the correct option.

A $Q = x^2 - 5x + 22, \quad R = -69x + 24$

B $Q = x^2 - 5x + 20, \quad R = -69x + 25$

C $Q = x^2 - 5x + 21, \quad R = -69x + 24$

D $Q = x^2 - 5x + 21, \quad R = 69x - 24$

Classified Practice

Polynomial Division

Q023 MCQ

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:

Choose the correct option.

A $Q = 3x + 2, \quad R = x + 6$

B $Q = 3x + 3, \quad R = x + 5$

C $Q = 3x + 3, \quad R = -x - 5$

D $Q = 3x + 4, \quad R = x + 5$

Q024 MCQ

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:

Choose the correct option.

A $Q = 2x^2 - 3x + 1, \quad R = 8x - 10$

B $Q = 2x^2 - 3x + 1, \quad R = -8x + 10$

C $Q = 2x^2 - 3x + 2, \quad R = 8x - 10$

D $Q = x \times (2x - 3), \quad R = 8x - 9$

Classified Practice

Polynomial Division

Q025 Written

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

Q026 Written**Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.**

Classified Practice

Polynomial Division

Q027 MCQ

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:

Choose the correct option.

A $Q = 2x - \frac{3}{2}, R = -\frac{35x}{2} + 9$

B $Q = 2x - \frac{1}{2}, R = \frac{35x}{2} - 9$

C $Q = 2x - \frac{5}{2}, R = \frac{35x}{2} - 8$

D $Q = 2x - \frac{3}{2}, R = \frac{35x}{2} - 9$

Q028 MCQ**Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:****Choose the correct option.**

A $Q = x - 1, \quad R = -4x + 3$

B $Q = x, \quad R = -4x + 2$

C $Q = x, \quad R = 4x - 2$

D $Q = x + 1, \quad R = -4x + 2$

Classified Practice

Polynomial Division

Q029 MCQ

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:

Choose the correct option.

- A $Q = \frac{5x}{2} - \frac{3}{4}, R = \frac{9x}{4} + \frac{21}{4}$
- B $Q = \frac{5x}{2} + \frac{1}{4}, R = -\frac{9x}{4} - \frac{21}{4}$
- C $Q = \frac{5x}{2} - \frac{7}{4}, R = -\frac{9x}{4} - \frac{17}{4}$
- D $Q = \frac{5x}{2} - \frac{3}{4}, R = -\frac{9x}{4} - \frac{21}{4}$

Classified Practice

Polynomial Division

Q030 MCQ

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:

Choose the correct option.

- A $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{67}{64}, R = \frac{37x}{64} + \frac{163}{32}$
- B $Q = \frac{3x^2}{4} + \frac{5x}{16} - \frac{61}{64}, R = \frac{37x}{64} + \frac{195}{32}$
- C $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, R = \frac{37x}{64} + \frac{163}{32}$
- D $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, R = -\frac{37x}{64} - \frac{163}{32}$

Classified Practice

Polynomial Division

Q031 MCQ

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:

Choose the correct option.

- A $Q = \frac{3x}{2} + \frac{11}{4}, R = \frac{11x}{4} - \frac{3}{4}$
- B $Q = \frac{3x}{2} + \frac{15}{4}, R = -\frac{11x}{4} + \frac{3}{4}$
- C $Q = \frac{3x}{2} + \frac{7}{4}, R = -\frac{11x}{4} + \frac{7}{4}$
- D $Q = \frac{3x}{2} + \frac{11}{4}, R = -\frac{11x}{4} + \frac{3}{4}$

Classified Practice

Polynomial Division

Q032 MCQ

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

Choose the correct option.

A $Q = 3x - \frac{11}{2}, \quad R = \frac{29x}{2} - 7$

B $Q = 3x - \frac{9}{2}, \quad R = \frac{29x}{2} - 8$

C $Q = 3x - \frac{9}{2}, \quad R = -\frac{29x}{2} + 8$

D $Q = 3x - \frac{7}{2}, \quad R = \frac{29x}{2} - 8$

Q033 Written

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q034 **Written**

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Q035 Written

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q036 Written

Find the quotient and the remainder when $-a^3 + a^2x + ax^2 + x^3$ is divided by $2a + x$. Treat it as a polynomial in x ; other letters are constants.

Q037 Written

Find the quotient and the remainder when $3a^3 - 2ax^2 + x^3$ is divided by $a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q038 MCQ

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x ; other letters are constants:

Choose the correct option.

A $Q = 4a^2 - 3ax + x^2, \quad R = -2a^3$

B $Q = 4a^2 - 3ax + x^2 + 1, \quad R = 2a^3$

C $Q = 4a^2 - 3ax + x^2 - 1, \quad R = 2a^3 + 1$

D $Q = 4a^2 - 3ax + x^2, \quad R = 2a^3$

Q039 MCQ

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants:

Choose the correct option.

A $Q = x + 2y - 1, \quad R = -6y^2 + 1$

B $Q = x + 2y, \quad R = -6y^2$

C $Q = x + 2y, \quad R = 6y^2$

D $Q = x + 2y + 1, \quad R = -6y^2$

Classified Practice

Polynomial Division

Q040 Written

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants.

Q041 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

Classified Practice

Polynomial Division

Q042 **Written**

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

Q043 Written

Find the polynomial A when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

Classified Practice

Polynomial Division

Q044 Written

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

Q045 Written

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

Classified Practice

Polynomial Division

Q046 Written

When $4x^3 + 8x - 9$ is divided by B , the quotient is $2x - 3$ and the remainder is $x + 15$. Find B .

Q047 Written

When $x^3 - x^2 + 3x + 1$ is divided by B , the quotient is $x^2 - 2x + 5$ and the remainder is -4 . Find B .

Classified Practice

Polynomial Division

Q048 **Written**

When $2x^3 + x^2 - 3x + 5$ is divided by B, the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B.

Q049 Written

When $6x^3 + 11x^2 - 2$ is divided by B, the quotient is $2x^2 + 3x - 1$ and the remainder is -1 . Find B.

Classified Practice

Polynomial Division

Q050 Written

When $2x^3 + 4x^2 + 7$ is divided by B , the quotient is $x + 2$ and the remainder is $3x + 13$. Find B .

Q051 Written

When $x^4 + x^2 + 3x - 1$ is divided by B , the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B .

Classified Practice

Polynomial Division

Q052 Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x .

Quiz MCQ 1 Concept Quiz · MCQ

For $x^2 - 5x + 1$ divided by $x - 2$, what is R?:

Choose the correct option.

A -5

B -1

C 5

D 1

Classified Practice

Polynomial Division

Quiz MCQ 2 **Concept Quiz · MCQ**

In a polynomial in x , a is treated as... Which of the following is correct?:

Choose the correct option.

A *another variable to divide by*

B a constant coefficient

C *zero*

D the divisor

Quiz MCQ 3 Concept Quiz · MCQ

To recover **B** from $A = B \times Q + R$, use... Which of the following is correct?:

Choose the correct option.

A $B = A \times Q + R$

B $B = \frac{A+R}{Q}$

C $B = \frac{A-R}{Q}$

D $B = \frac{Q}{A-R}$

Classified Practice

Polynomial Division

Quiz MCQ 4 **Concept Quiz · MCQ**

If B has degree 2, a valid remainder can have degree...:

Choose the correct option.

A 3

B 2

C 1

D 4

Quiz WRITTEN 5 Concept Quiz · Written

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q , R , and verify $A = B \times Q + R$.

Classified Practice

Polynomial Division

Quiz WRITTEN 6 Concept Quiz · Written**Recover A when $B=x^2 - x - 2$, $Q=2x + 3$, $R=-x + 1$.**

Quiz WRITTEN 7 Concept Quiz · Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

Classified Practice

Polynomial Division

Quiz WRITTEN 8 Concept Quiz · Written

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

Answer key

Use the question number shown on each page.

Q001 B	Q016 $Q = 2x - 4$, $R = x - 11$	Q031 D	Q046 $2x^2 + 3x + 8$
Q002 A	Q017 $Q = 2x + 1$, $R = -x + 1$	Q032 B	Q047 $x + 1$
Q003 D	Q018 $Q = x - 2$, $R = 0$	Q033 $Q = 3a + 4x$, $R = 0$	Q048 $2x^2 - x + 3$
Q004 C	Q019 D	Q034 $Q = 3a^2 - ax + x^2$, $R = a^3$	Q049 $3x + 1$
Q005 B	Q020 B	Q035 $Q = 3a + 4x$, $R = 0$	Q050 $2x^2 - 3$
Q006 A	Q021 D	Q036 $Q = 3a^2 - ax + x^2$, $R = -7a^3$	Q051 $x^2 - x + 1$
Q007 $Q = x - 3$, $R = -5$	Q022 C	Q037 $Q = 3a^2 - 3ax + x^2$, $R = 0$	Q052 $2x + 3$
Q008 $Q = x^2 + 3x + 5$, $R = 16$	Q023 B	Q038 D	Quiz MCQ 1 A
Q009 D	Q024 A	Q039 B	Quiz MCQ 2 B
Q010 B	Q025 $Q = 3x - 1$, $R = 13x$	Q040 $Q = x^2 + 2xy - 2y^2$, $R = 0$	Quiz MCQ 3 C
Q011 B	Q026 $Q = -3x^2 - 4x - 18$, $R = 63x + 35$	Q041 $x^3 + x^2 - 9x + 2$	Quiz MCQ 4 C
Q012 A	Q027 D	Q042 $2x^3 - 7x^2 + 2x + 3$	Quiz WRITTEN 5 $Q = 3x^2 + x + 4$, $R = 1$
Q013 D	Q028 B	Q043 $2x^3 - 12x + 9$	Quiz WRITTEN 6 $2x^3 + x^2 - 8x - 5$
Q014 B	Q029 D	Q044 $4x^3 - 4x^2 + 5x$	Quiz WRITTEN 7 $2x + 3$
Q015 $Q = x - 2$, $R = -x - 4$	Q030 C	Q045 $6x^3 + 7x^2 + 3x + 8$	Quiz WRITTEN 8 $Q = x^2 - x - 2$, $R = 2x + 1$